

Exercise of Supervised Learning: SVM Part 2

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Exercise 1: Kernelized Multiclass SVM

For a data set $\mathcal{D} = \{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})\}$ with $y^{(i)} \in \mathcal{Y} = \{+1, -1\}$, assume that we are provided with a suitable feature map $\phi : \mathcal{X} \rightarrow \Phi$, where $\Phi \subset \mathbb{R}^d$. In the featureized SVM learning problem we are facing the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} & \frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} + C \sum_{i=1}^n \zeta^{(i)} \\ \text{s.t. } & y^{(i)} \left(\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)}) \rangle + \theta_0 \right) \geq 1 - \zeta^{(i)} \quad \forall i \in \{1, \dots, n\}, \\ & \text{and } \zeta^{(i)} \geq 0 \quad i \in \{1, \dots, n\}, \end{aligned}$$

where $C \geq 0$ is some constant.

(a) Argue that this is equivalent to the following ERM problem:

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2} \|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^n \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0).$$

i.e., the regularized ERM problem for the hinge loss for the hypothesis space

$$\mathcal{H} = \{f : \Phi \rightarrow \mathbb{R} \mid f(\mathbf{z}) = \boldsymbol{\theta}^\top \mathbf{z} + \theta_0, \boldsymbol{\theta} \in \mathbb{R}^d, \theta_0 \in \mathbb{R}\}$$

1(a): Rewriting the Optimization Target

Optimization target:

$$\begin{aligned} \min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} & \frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} + c \sum_{i=1}^n \zeta^{(i)} \\ \text{s.t. } & \zeta^{(i)} \geq 1 - y^{(i)} \left(\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)}) \rangle + \theta_0 \right), \quad \forall i, \\ \text{and } & \zeta^{(i)} \geq 0, \quad \forall i. \end{aligned}$$

1(a): Comparison between Optimization Target and $\mathcal{R}_{\text{emp}}$

Optimization target:

$$\begin{aligned} \min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} \quad & \frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} + C \sum_{i=1}^n \zeta^{(i)} \\ \text{s.t.} \quad & \zeta^{(i)} \geq 1 - y^{(i)} \left(\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)}) \rangle + \theta_0 \right), \quad \forall i, \\ & \text{and } \zeta^{(i)} \geq 0, \quad \forall i. \end{aligned}$$

Empirical risk:

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2} \|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^n \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0).$$

Observation: Both contain $\frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta}$. Both contain $C \sum_{i=1}^n \dots$ penalty terms.

Next: Prove that $\zeta^{(i)}$ equals $\max(\dots)$ term.

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Next: Prove that $\zeta^{(i)}$ equals $\max(\dots)$ term.

1 (a): Prove $\zeta^{(i)}$ Equals $\max(\dots)$ Term

For each i , The constraints in the optimization problem:

$$\zeta^{(i)} \geq \underbrace{1 - y^{(i)} (\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)}) \rangle + \theta_0)}_{(i)}$$

$$\zeta^{(i)} \geq \underbrace{0}_{(ii)}$$

$\zeta^{(i)} \geq (i) \text{ and } (ii) \Rightarrow \zeta^{(i)} \geq \text{the larger term in (i) and (ii)} \Rightarrow \zeta^{(i)} \geq \max((i), (ii)).$

Therefore, the constraints translate to $\zeta^{(i)} \geq \max(1 - y^{(i)} (\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0)$

Note: Our target is to $\min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} \frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} + C \sum_{i=1}^n \zeta^{(i)}$, so smaller $\zeta^{(i)}$ are preferred. $\rightsquigarrow$ Choose $\zeta^{(i)} = \max(\dots, 0)$

recall SVM Part 1, Ex. 1(b)

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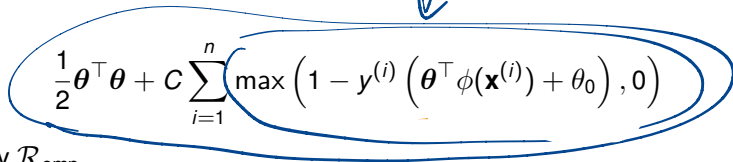
recall SVM Part 1, Ex. 1(b)

1 (a): Choose $\zeta^{(i)} = \max(\dots, 0)$

The optimization target

$$\frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} + c \sum_{i=1}^n \zeta^{(i)}$$


becomes


$$\frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} + c \sum_{i=1}^n \max \left(1 - y^{(i)} \left(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0 \right), 0 \right)$$

which is exactly $\mathcal{R}_{\text{emp}}$.



Exercise 1 (b)

$$\theta^T \phi(x^{(i)}) + \theta_0 = \begin{pmatrix} \theta \\ \theta_0 \end{pmatrix}^T \begin{pmatrix} \phi(x^{(i)}) \\ 1 \end{pmatrix}$$

(b) Now assume we deal with a multiclass classification problem with a data set $\mathcal{D} = \{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})\}$ such that $y^{(i)} \in \mathcal{Y} = \{1, \dots, g\}$ for each $i \in \{1, \dots, n\}$. In this case, we can derive a similar regularized ERM problem by using the multiclass hinge loss (see Exercise Sheet 4(b)):

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2} \|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^n \sum_{y \neq y^{(i)}} \max(1 + \tilde{\boldsymbol{\theta}}^T \psi(\mathbf{x}^{(i)}, y) - \tilde{\boldsymbol{\theta}}^T \psi(\mathbf{x}^{(i)}, y^{(i)}), 0),$$

where $\tilde{\boldsymbol{\theta}} := (\theta_0, \boldsymbol{\theta}^T)^T \in \mathbb{R}^{d+1}$, and $\psi : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^{d+1}$ is a suitable (multiclass) feature map. Specify a ψ such that this regularized multiclass ERM problem coincides with the regularized binary ERM problem in (a).

1 (b): Think Before Solving

(a) binary: $\mathcal{R}_{\text{emp}}(\theta) = \frac{1}{2}\|\theta\|^2 + C \sum_{i=1}^n \max(1 - y^{(i)}(\theta^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0)$

(b) multiclass: $\mathcal{R}_{\text{emp}}(\theta) = \frac{1}{2}\|\theta\|^2 + C \sum_{i=1}^n \sum_{y \neq y^{(i)}} \max(1 + \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y) - \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y^{(i)}), 0)$

-
- **Multiclass.** SVM is binary by design — one-vs-one, one-vs-rest, or one joint loss?
 - **Intercept.** Margin needs $+\theta_0$, but ψ gives only an inner product $\langle \tilde{\theta}, \psi(\mathbf{x}^{(i)}, y) \rangle$. How to absorb θ_0 ?
 - **Equivalence.** Why do (a) and (b) coincide in the binary case ($g = 2$)?

1 (b): Think Before Solving

$$(a) \text{ binary: } \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2} \|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^n \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \boldsymbol{\phi}(\mathbf{x}^{(i)})) + \theta_0, 0)$$

$$(b) \text{ multiclass: } \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2} \|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^n \sum_{y \neq y^{(i)}} \max(1 + \tilde{\boldsymbol{\theta}}^\top \boldsymbol{\psi}(\mathbf{x}^{(i)}, y) - \tilde{\boldsymbol{\theta}}^\top \boldsymbol{\psi}(\mathbf{x}^{(i)}, y^{(i)}), 0)$$

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1(b): A Closer Look at the Multiclass Hinge Loss

Goal: multiclass hinge $\rightarrow$ binary hinge when $g = 2$.

- ▶ Binary encoding: $y^{(i)} \in \{1, 2\} \mapsto \{-1, +1\}$.
- ▶ Then $\sum_{y \neq y^{(i)}}$ collapses to **one term** ($y = -y^{(i)}$).
- ▶ $\psi(\mathbf{x}^{(i)}, y^{(i)})$ uses *both* $\mathbf{x}^{(i)}$ and $y^{(i)}$ — unlike $\phi(\mathbf{x}^{(i)})$.
- ▶ No θ_0 in the multiclass loss $\Rightarrow$ absorb it how? (*next*)

Roadmap

▶ Multiclass

one joint hinge loss

Intercept (θ_0)

dummy feature 1

Equivalence

$g=2 \Rightarrow$ binary (a)

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1(b): A Closer Look at the Multiclass Hinge Loss

$$g = 5$$
$$Y = \{1, 2\}$$

Goal: multiclass hinge $\rightarrow$ binary hinge when $g = 2$.

► Binary encoding: $y^{(i)} \in \{1, 2\} \mapsto \{-1, +1\}$.

$$y^{(i)} = -1$$

► Then $\sum_{y \neq y^{(i)}}$ collapses to **one term** ($y = -y^{(i)}$).

► $\psi(\mathbf{x}^{(i)}, y^{(i)})$ uses *both* $\mathbf{x}^{(i)}$ and $y^{(i)}$ — unlike $\phi(\mathbf{x}^{(i)})$.

► No θ_0 in the multiclass loss $\Rightarrow$ absorb it how? (next)

$$y \neq y^{(i)} \rightarrow \begin{cases} y^{(i)} = -1 \\ y^{(i)} = +1 \end{cases}$$

$$y \rightarrow \{-1, +1\}$$

Roadmap

► **Multiclass**

one joint hinge loss

Intercept (θ_0)

dummy feature 1

Equivalence

$g=2 \Rightarrow$ binary (a)

$$y = +1$$
$$y = -1$$

1(b): Define $\psi : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^{d+1}$

Goal: write the margin $y(\theta^\top \phi(\mathbf{x}) + \theta_0)$ as one inner product.

1. Absorb θ_0 via a dummy feature 1:

$$\theta^\top \phi(\mathbf{x}) + \theta_0 = \underbrace{\begin{pmatrix} \theta_0 \\ \theta \end{pmatrix}^\top}_{\tilde{\theta}^\top} \underbrace{\begin{pmatrix} 1 \\ \phi(\mathbf{x}) \end{pmatrix}}_{\tilde{\phi}(\mathbf{x})}$$

2. Pull in y : $y \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}) = \tilde{\theta}^\top (y \tilde{\phi}(\mathbf{x}))$

3. Scale by $\frac{1}{2}$:

$$\psi(\mathbf{x}, y) = \frac{1}{2} y \tilde{\phi}(\mathbf{x})$$

Roadmap

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one joint hinge loss

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$$\tilde{\theta}^\top \tilde{\phi}(\mathbf{x}) \cdot y$$

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Roadmap

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one joint hinge loss

► Intercept (θ_0)

dummy feature 1

Equivalence

$g=2 \Rightarrow$ binary (a)

1 (b): What We Prove Next

$$\sum_{y \neq y^{(i)}} \dots = \text{sum over one class}$$

With $\psi(\mathbf{x}, y) = \frac{1}{2} y \tilde{\phi}(\mathbf{x})$, it remains to show — in the binary case
($g = 2$) — that the multiclass penalty reduces to the binary hinge of (a):

$$\underbrace{\sum_{y \neq y^{(i)}} \max(1 + \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y) - \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y^{(i)}), 0)}_{\text{(b) multiclass}}$$
$$= \underbrace{\max(1 - y^{(i)}(\theta^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0)}_{\text{(a) binary}}.$$

Roadmap

✓ **Multiclass**

one joint hinge loss

✓ **Intercept (θ_0)**

dummy feature 1

► **Equivalence**

$g=2 \Rightarrow$ binary (a)

1 (b): Reduce to the Binary Hinge

Binary case $\Rightarrow$ the sum has **one term** $(y = -y^{(i)})$ Then:

$$\begin{aligned} & 1 + \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y) - \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y^{(i)}) \\ &= 1 + \frac{1}{2} y \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}) - \frac{1}{2} y^{(i)} \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}) \\ &= 1 + \frac{1}{2} (y - y^{(i)}) \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}) \\ &= \begin{cases} 1 + \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}), & y^{(i)} = -1 \\ 1 - \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}), & y^{(i)} = +1 \end{cases} \\ &= 1 - y^{(i)} \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}). \end{aligned}$$

$$\sum_{y \neq y^{(i)}}$$

$\Rightarrow$ Hinge loss on top of $y = -y^{(i)}$

Roadmap

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one joint hinge loss

✓ **Intercept** (θ_0)

dummy feature 1

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$g=2 \Rightarrow$ binary (a)

1 (b): Reduce to the Binary Hinge

$$\psi(x^{(i)}, y^{(i)}) = \frac{1}{2} y \tilde{\phi}(x^{(i)})$$

Binary case $\Rightarrow$ the sum has **one term** ($y = -y^{(i)}$). Then:

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Roadmap

✓ **Multiclass**

one joint hinge loss

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dummy feature 1

► **Equivalence**

$g=2 \Rightarrow$ binary (a)

Solution to 1 (b): Continued

From the per-term reduction, the sum collapses to a single term:

$$\begin{aligned}\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) &= \frac{1}{2}\|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^n \max(1 - y^{(i)} \tilde{\boldsymbol{\theta}}^\top \tilde{\boldsymbol{\phi}}(\mathbf{x}^{(i)}), 0) \\ &= \frac{1}{2}\|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^n \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \boldsymbol{\phi}(\mathbf{x}^{(i)}) + \theta_0), 0).\end{aligned}$$

multiclass ERM = binary ERM (a)



Roadmap

✓ **Multiclass**

one joint hinge loss

✓ **Intercept (θ_0)**

dummy feature 1

► **Equivalence**

$g=2 \Rightarrow$ binary (a)

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Roadmap

✓ **Multiclass**

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► **Equivalence**

$g=2 \Rightarrow$ binary (a)

Exercise 1 (c)

$$K \in \mathbb{R}^{ng \times ng}$$

Show that the regularized multiclass ERM problem in (b) can be written in the kernelized form:

$$\frac{1}{2} \beta^\top K \beta + C \sum_{i=1}^n \sum_{y \neq y^{(i)}} \max(1 + (K\beta)_{(i-1)g+y} - (K\beta)_{(i-1)g+y^{(i)}}, 0),$$

where $\beta \in \mathbb{R}^{ng}$ and $K = \mathbf{X}\mathbf{X}^\top$ for $\mathbf{X} \in \mathbb{R}^{ng \times (d+1)}$ with row entries $\psi(\mathbf{x}^{(i)}, y)^\top$ for $i = 1, \dots, n$, $y = 1, \dots, g$, i.e.,

$$\mathbf{X} = \begin{pmatrix} \psi(\mathbf{x}^{(1)}, 1)^\top \\ \psi(\mathbf{x}^{(1)}, 2)^\top \\ \vdots \\ \psi(\mathbf{x}^{(1)}, g)^\top \\ \psi(\mathbf{x}^{(2)}, 1)^\top \\ \vdots \\ \psi(\mathbf{x}^{(n)}, g)^\top \end{pmatrix} \quad \left. \vphantom{\begin{pmatrix} \psi(\mathbf{x}^{(1)}, 1)^\top \\ \psi(\mathbf{x}^{(1)}, 2)^\top \\ \vdots \\ \psi(\mathbf{x}^{(1)}, g)^\top \\ \psi(\mathbf{x}^{(2)}, 1)^\top \\ \vdots \\ \psi(\mathbf{x}^{(n)}, g)^\top \end{pmatrix}} \right\} \text{ from } \mathcal{X}^{(d)}$$

Here, $(K\beta)_{(i-1)g+y}$ is the $((i-1)g+y)$ -th entry of $K\beta$.

How to get there?

1. Representer theorem: $\theta^* \in \text{span}\{\psi(\mathbf{x}^{(i)}, y)\}$.
Rewrite θ with $\mathbf{X}$ and β ?

2. Then express $\frac{1}{2} \|\theta\|^2$ via $K = \mathbf{X}\mathbf{X}^\top$ and β ?

3. And the scores $\theta^\top \psi(\mathbf{x}^{(i)}, y)$ via $K\beta$?

$$\theta = \beta_1 \psi(\mathbf{x}^{(1)}, 1) + \beta_2 \psi(\mathbf{x}^{(1)}, 2) + \dots + \beta_{ng} \psi(\mathbf{x}^{(n)}, g)$$

1 (c): Express $\|\theta\|^2$ with K and β

Representer theorem: θ^* is a linear combination of the $\psi(\mathbf{x}^{(i)}, y)$, i.e.

$$\theta = \mathbf{X}^\top \beta, \quad \beta \in \mathbb{R}^{ng},$$

with the $\psi(\mathbf{x}^{(i)}, y)^\top$ as the rows of $\mathbf{X}$ (as on the previous slide). With

$$K = \mathbf{X}\mathbf{X}^\top:$$

$$\|\theta\|^2 = \theta^\top \theta = (\mathbf{X}^\top \beta)^\top \mathbf{X}^\top \beta = \boxed{\beta^\top K \beta}.$$

Roadmap

✓ Represent

$$\theta = \mathbf{X}^\top \beta$$

► Norm

$$\|\theta\|^2 = \beta^\top K \beta$$

Scores

$$\theta^\top \psi \rightarrow (K\beta)$$

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$$\|\theta\|^2 = \theta^\top \theta = \underbrace{(\mathbf{X}^\top \beta)^\top}_{\text{}} \underbrace{\mathbf{X}^\top \beta}_{\text{}} = \boxed{\beta^\top K \beta}.$$

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$$\theta = \mathbf{X}^\top \beta$$

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$$\|\theta\|^2 = \beta^\top K \beta$$

Scores

$$\theta^\top \psi \rightarrow (K\beta)$$

1 (c): Express $\theta^\top \psi(\mathbf{x}^{(i)}, y)$ with $\mathbf{K}$ and β

Since $\theta = \mathbf{X}^\top \beta$: $\theta^\top \psi(\mathbf{x}^{(i)}, y) = \beta^\top \mathbf{X} \psi(\mathbf{x}^{(i)}, y)$ (scalar).

$$\left. \begin{array}{c} (i-1)g \\ \text{rows} \end{array} \right\} \left[\begin{array}{c} \psi(\mathbf{x}^{(1)}, 1)^\top \\ \vdots \\ \psi(\mathbf{x}^{(i-1)}, g)^\top \\ \psi(\mathbf{x}^{(i)}, 1)^\top \\ \psi(\mathbf{x}^{(i)}, y)^\top \\ \psi(\mathbf{x}^{(i)}, g)^\top \end{array} \right] \left. \vphantom{\begin{array}{c} \psi(\mathbf{x}^{(i)}, y)^\top \end{array}} \right\} \begin{array}{c} \text{block } i \\ (g \text{ rows}) \end{array}$$

So $\psi(\mathbf{x}^{(i)}, y)$ is row $(i-1)g+y$ of $\mathbf{X}$, i.e. column $(i-1)g+y$ of $\mathbf{X}^\top$.

Next: form $\mathbf{X} \psi(\mathbf{x}^{(i)}, y)$ via $\mathbf{K} = \mathbf{X} \mathbf{X}^\top$.

Roadmap

✓ Represent

$$\theta = \mathbf{X}^\top \beta$$

✓ Norm

$$\|\theta\|^2 = \beta^\top \mathbf{K} \beta$$

► Scores

$$\theta^\top \psi \rightarrow (\mathbf{K} \beta)$$

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Since $\theta = \mathbf{X}^\top \beta$: $\theta^\top \psi(\mathbf{x}^{(i)}, y) = \beta^\top \mathbf{X} \psi(\mathbf{x}^{(i)}, y)$ (scalar).

$$(i-1)g \left\{ \begin{array}{c} \psi(\mathbf{x}^{(1)}, 1)^\top \\ \vdots \\ \psi(\mathbf{x}^{(i-1)}, g)^\top \\ \psi(\mathbf{x}^{(i)}, 1)^\top \\ \psi(\mathbf{x}^{(i)}, y)^\top \\ \psi(\mathbf{x}^{(i)}, g)^\top \end{array} \right\} \left. \begin{array}{c} \\ \\ \\ \\ \text{block } i \\ (g \text{ rows}) \end{array} \right\}$$

So $\psi(\mathbf{x}^{(i)}, y)$ is row $(i-1)g+y$ of $\mathbf{X}$, i.e. column $(i-1)g+y$ of $\mathbf{X}^\top$.

Next: form $\mathbf{X} \psi(\mathbf{x}^{(i)}, y)$ via $K = \mathbf{X} \mathbf{X}^\top$.

Roadmap

✓ Represent

$$\theta = \mathbf{X}^\top \beta$$

✓ Norm

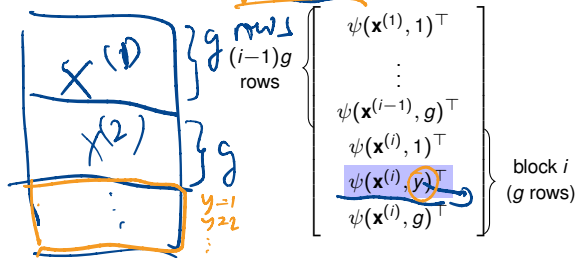
$$\|\theta\|^2 = \beta^\top K \beta$$

► Scores

$$\theta^\top \psi \rightarrow (K\beta)$$

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Since $\theta = \mathbf{X}^\top \beta$: $\theta^\top \psi(\mathbf{x}^{(i)}, y) = \beta^\top \mathbf{X} \psi(\mathbf{x}^{(i)}, y)$ (scalar).



So $\psi(\mathbf{x}^{(i)}, y)$ is row $(i-1)g + y$ of $\mathbf{X}$, i.e. column $(i-1)g + y$ of $\mathbf{X}^\top$.

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Roadmap

✓ Represent

$$\theta = \mathbf{X}^\top \beta$$

✓ Norm

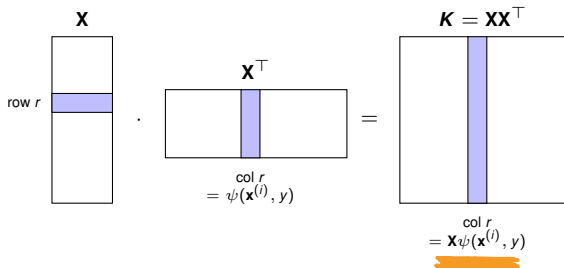
$$\|\theta\|^2 = \beta^\top \mathbf{K} \beta$$

► Scores

$$\theta^\top \psi \rightarrow (\mathbf{K} \beta)$$

1 (c): $\mathbf{X} \psi(\mathbf{x}^{(i)}, y)$ is a column of $\mathbf{K} = \mathbf{X} \mathbf{X}^\top$

With $r := (i-1)g+y$: $\psi(\mathbf{x}^{(i)}, y) = \text{col } r \text{ of } \mathbf{X}^\top$, so $\mathbf{X} \psi(\mathbf{x}^{(i)}, y)$ is column r of $\mathbf{X} \mathbf{X}^\top$:



$\Rightarrow \mathbf{X} \psi(\mathbf{x}^{(i)}, y) = \text{col } r \text{ of } \mathbf{K} (= \text{row } r, \text{ symmetric})$

$\Rightarrow \beta^\top \mathbf{X} \psi(\mathbf{x}^{(i)}, y) = (\mathbf{K} \beta)_r$.

$$\mathcal{R}_{\text{emp}}(\theta) = \frac{1}{2} \beta^\top \mathbf{K} \beta + \sum_{i=1}^n \sum_{y \neq y^{(i)}} \max(1 + (\mathbf{K} \beta)_{(i-1)g+y} - (\mathbf{K} \beta)_{(i-1)g+y^{(i)}}, 0).$$

Roadmap

✓ Represent

$$\theta = \mathbf{X}^\top \beta$$

✓ Norm

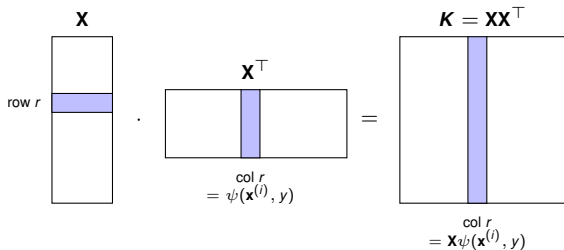
$$\|\theta\|^2 = \beta^\top \mathbf{K} \beta$$

► Scores

$$\theta^\top \psi \rightarrow (\mathbf{K} \beta)$$

1 (c): $\mathbf{X}\psi(\mathbf{x}^{(i)}, y)$ is a column of $\mathbf{K}$

With $r := (i-1)g+y$: $\psi(\mathbf{x}^{(i)}, y) = \text{col } r$ of $\mathbf{X}^\top$, so $\mathbf{X}\psi(\mathbf{x}^{(i)}, y)$ is column r of $\mathbf{X}\mathbf{X}^\top$:



Roadmap

✓ Represent

$$\theta = \mathbf{X}^\top \beta$$

✓ Norm

$$\|\theta\|^2 = \beta^\top \mathbf{K} \beta$$

► Scores

$$\theta^\top \psi \rightarrow (\mathbf{K}\beta)$$

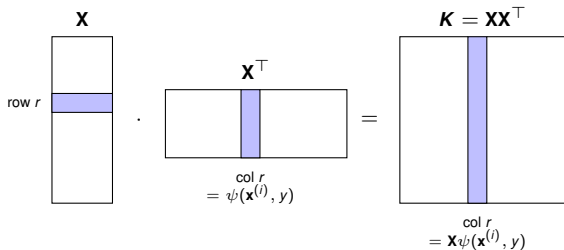
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$$\Rightarrow \beta^\top \mathbf{X}\psi(\mathbf{x}^{(i)}, y) = (\mathbf{K}\beta)_r.$$

$$\mathcal{R}_{\text{emp}}(\theta) = \frac{1}{2} \beta^\top \mathbf{K} \beta + \sum_{i=1}^n \sum_{y \neq y^{(i)}} \max(1 + (\mathbf{K}\beta)_{(i-1)g+y} - (\mathbf{K}\beta)_{(i-1)g+y^{(i)}}, 0).$$

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Exercise 2: Kernel Trick

$$K(\cdot, \cdot) \xrightarrow[\text{Gram matrix}]{\text{induces}} K \longrightarrow \text{P.S.D.}$$

The polynomial kernel is defined as

$$k(x, \tilde{x}) = (x^\top \tilde{x} + b)^d.$$

Furthermore, assume that $x \in \mathbb{R}^2$ and $d = 2$. (a) Derive the explicit feature map ϕ taking into account that the following equation holds:

$$k(x, \tilde{x}) = \langle \phi(x), \phi(\tilde{x}) \rangle$$

$$K_{ij} = k(x^{(i)}, x^{(j)})$$
$$\downarrow$$
$$K_{ij} = K_{ji}$$

Solution to 2 (a)

$$k(x, \tilde{x}) = (x^\top \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2$$


Solution to 2 (a)

$$\begin{aligned}k(x, \tilde{x}) &= (x^\top \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2 \\&= (x_1 \tilde{x}_1 + x_2 \tilde{x}_2 + b)^2\end{aligned}$$

Solution to 2 (a)

① $\rightarrow$ element $\dots \begin{pmatrix} \vdots \\ \vdots \end{pmatrix}$

$$\phi(x)^T \phi(\tilde{x}) = \langle \phi(x), \phi(\tilde{x}) \rangle$$

$x^T \tilde{x} \rightarrow$ scalar

$$k(x, \tilde{x}) = (x^T \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^T \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2$$

$$= (x_1 \tilde{x}_1 + x_2 \tilde{x}_2 + b)^2$$

$$= x_1^2 \tilde{x}_1^2 + 2x_1 \tilde{x}_1 x_2 \tilde{x}_2 + x_2^2 \tilde{x}_2^2 + 2bx_1 \tilde{x}_1 + 2bx_2 \tilde{x}_2 + b^2$$

x $\begin{pmatrix} \vdots \\ \vdots \end{pmatrix}^T \cdot \begin{pmatrix} \vdots \\ \vdots \end{pmatrix}$ $\tilde{x}$

18/20

Solution to 2 (a)

$$k(x, \tilde{x}) = (x^\top \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2$$

$\frac{1 \cdot (1 - (x^\top \tilde{x})^n)}{1 - x^\top \tilde{x}} \xrightarrow{n \rightarrow \infty} \frac{1}{1 - x^\top \tilde{x}}$
 $x, \tilde{x} \in (0, 1)$

$$v^\top | \langle v | v \rangle > 0$$

$$= (x_1 \tilde{x}_1 + x_2 \tilde{x}_2 + b)^2$$

$$= x_1^2 \tilde{x}_1^2 + 2x_1 \tilde{x}_1 x_2 \tilde{x}_2 + x_2^2 \tilde{x}_2^2 + 2bx_1 \tilde{x}_1 + 2bx_2 \tilde{x}_2 + b^2$$

$\uparrow \quad \uparrow \quad = b \cdot b$

$$= \left\langle \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \\ \sqrt{2}bx_1 \\ \sqrt{2}bx_2 \\ b \end{pmatrix}, \begin{pmatrix} \tilde{x}_1^2 \\ \sqrt{2}\tilde{x}_1\tilde{x}_2 \\ \tilde{x}_2^2 \\ \sqrt{2}b\tilde{x}_1 \\ \sqrt{2}b\tilde{x}_2 \\ b \end{pmatrix} \right\rangle$$

$2 = \sqrt{2} \cdot \sqrt{2}$

$$= \langle \phi(x), \phi(\tilde{x}) \rangle$$

Exercise 2 (b)

(b) Describe the main differences between the kernel method and the explicit feature map.

Solution: Using the kernel method reduces the computational cost of the scalar product in the higher-dimensional feature space, since the feature map is never formed explicitly.

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Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
1 (a)	Featurized soft-margin SVM = regularized ERM with the hinge loss: $\frac{1}{2} \ \boldsymbol{\theta}\ ^2 + C \sum_{i=1}^n \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)} + \boldsymbol{\theta}_0), 0)$.	Hinge loss; SVM as regularized empirical risk minimization.
1 (b)	Joint feature map $\psi(\mathbf{x}, y) = \frac{1}{2} y \tilde{\phi}(\mathbf{x})$, $\tilde{\phi} = (1, \phi)^\top$ (absorbs $\boldsymbol{\theta}_0$); the multiclass hinge collapses to the binary hinge when $g = 2$.	Multiclass SVM via a joint feature map $\psi(\mathbf{x}, y)$; multiclass hinge loss.
1 (c)	Representer theorem $\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta} \Rightarrow$ kernelized form $\frac{1}{2} \boldsymbol{\beta}^\top \mathbf{K} \boldsymbol{\beta} + \sum_{i=1}^n \sum_{y \neq y^{(i)}} \max(1 + (\mathbf{K} \boldsymbol{\beta})_{(i-1)g+y} - (\mathbf{K} \boldsymbol{\beta})_{(i-1)g+y^{(i)}}, 0)$, $\mathbf{K} = \mathbf{X} \mathbf{X}^\top$.	Representer theorem; kernelization; Gram matrix $\mathbf{K}$.
2	Polynomial kernel $(\mathbf{x}^\top \tilde{\mathbf{x}} + b)^2 = \langle \phi(\mathbf{x}), \phi(\tilde{\mathbf{x}}) \rangle$ with explicit $\phi \in \mathbb{R}^6$; the kernel returns the inner product without forming ϕ .	Kernel trick; kernels vs. explicit feature maps (compute savings).